

xStore

Analytics & Business Intelligence Requirements

What the app and admin dashboard need to measure the business —
and what's missing today.

Prepared for: Backend Team · Stakeholders · Marketing · Vendor Ops · Delivery Ops

Prepared by: Mobile Team (Business Review)

Date: 17 August 2026 · Version 1.0

Contents

1. Executive Summary
2. Current State — What's Built vs. What's Missing
3. Business-Critical Gaps
4. Requirements by Audience
5. Consolidated Event Catalog
6. Business Flows
7. Dashboard Features Required
8. Prioritized Roadmap
9. Decisions Needed From the Owner
10. Source Reference

1. Executive Summary

xStore is a cash-on-delivery (COD) marketplace for the Egyptian market, connecting consumers with vendors, plus an emerging “Send Package” peer-to-peer delivery line. Two things drive revenue today: **GMV from COD orders** and the **flat 2% commission** collected from vendors on those orders. A third, the delivery pilot, is a live but completely unmeasured revenue line.

You cannot manage what you cannot see. The mobile app already ships a custom analytics pipeline covering the core buyer funnel (view → cart → checkout → purchase) — that work is solid and should not be redone. But this review found the pipeline stops exactly where three real revenue and risk questions begin: *is vendor commission being collected, is the delivery pilot worth the investment, and where in the funnel are we actually losing customers?* None of those are answerable from what's instrumented today.

This document is a business requirements brief, written for the backend team to build against and for stakeholders / marketing / vendor ops / delivery ops to know what they will receive and when. It does three things: (1) states what's already built so nobody re-specs it, (2) names the gaps that carry real business risk in plain terms, and (3) hands each team a concrete list scoped to their function.

The one-sentence version

The buyer funnel is instrumented; the money (commission) and the newest revenue line (delivery) are not — and the backend collector that turns any of this into a dashboard hasn't been built yet.

2. Current State — What's Built vs. What's Missing

Reviewed directly against the codebase and the existing `docs_business/` handoff notes.

Component	Status	Detail
Mobile event pipeline	■ Built & shipping	Batches locally, retries with backoff, auto screen_view tracking. <code>lib/core/analytics/analytics_service.dart</code>
Event catalog	■ 16 events frozen	Covers auth, browse, cart, checkout, purchase, wishlist, listings, orders. <code>event_names.dart</code>
Backend collector POST <code>/api/analytics/events</code>	■ Not built	Spec is written and ready to implement. Every event fired since launch is queued client-side, going nowhere.
Admin dashboard UI	■ Prototype only	Static HTML/CSS/JS, all data hard-coded. No live backend wiring.
Admin/vendor REST API	■ Not built	Full endpoint spec exists (orders, vendors, couriers, commission, disputes) — ready for backend to build against.
Funnel KPI reporting	■ No live report	No dashboard consumes the events yet — funnel numbers exist nowhere for anyone to look at today.
Commission wallet visibility	■ Not tracked	Thresholds and settlement exist in the data model; no event marks when they fire.
Delivery pilot visibility	■ Not tracked	Zero analytics on the “Send Package” flow end to end.

Read this as: the client-side foundation is genuinely good — do not re-architect it. The gap is entirely on the receiving end (backend collector, dashboard) and in the surface area it doesn't yet cover (commission, delivery, drop-off diagnostics).

3. Business-Critical Gaps

Ranked by revenue/risk exposure, not by engineering effort. Each of these leaves a real business question unanswerable today.

■ Vendor commission — invisible revenue leakage

Business risk: The 2% commission is xStore's only vendor-side revenue lever today, and there is no event marking when a vendor crosses the warn/pause threshold or settles a balance. Commission revenue trend, at-risk vendors, and collection reliability are invisible without a manual database query.

Fix: Instrument *commission_threshold_warned*, *commission_paused*, *commission_settled*; surface commission KPIs on the admin home screen.

■ “Send Package” delivery pilot — a full revenue line with zero telemetry

Business risk: This is a second, distinct revenue line (admin-priced, cash collected at pickup) sharing the courier cash ledger with COD orders — and it has no analytics at all. There is no way to see where the submitted → priced → confirmed → delivered funnel leaks, or whether the pilot is worth further investment.

Fix: Instrument the full lifecycle plus *courier_cash_handover* so the pilot can be evaluated on its own funnel, not guessed at.

■ Cart abandonment & checkout failure — the highest-value diagnostic is missing

Business risk: The funnel only captures the *successful* path (view → cart → checkout → purchase). There is no *cart_remove* or *checkout_failed* event, so when conversion drops, nobody can tell whether it's cart abandonment, an address problem, or a failed COD confirmation. This is the single cheapest fix with the highest diagnostic value for the app's own stated north-star funnel.

Fix: Add *view_cart*, *cart_remove*, *checkout_failed* / *payment_declined*.

■ Coupons — marketing spend flying blind

Business risk: Coupon apply/remove happens inside checkout with zero tracking. Marketing cannot measure promo effectiveness, redemption rate, or discount cost against the incremental GMV it produced.

Fix: Add *coupon_applied* / *coupon_removed* / *coupon_failed* with code and discount value as properties.

■ Governorate / city selection — geography decisions are blind

Business risk: COD delivery risk and cost vary heavily by governorate in Egypt. No event captures the city/governorate chosen at checkout, so nobody can cohort conversion, AOV, or cancellation by geography — exactly the data needed to decide where to expand vendor supply or restrict COD.

Fix: Add *address_location_selected* (governorateId, cityId); tag *purchase* with delivery city.

■ Push notification engagement — a cost center with no ROI signal

Business risk: Notifications drive order updates and re-engagement, but open/tap events don't exist, so notification ROI and channel effectiveness are unmeasured spend.

Fix: Add *notification_received* / *notification_opened*.

■ Order data-model mismatch — flag for backend before trusting GMV numbers

Business risk: The orders API is documented as one listing per order, while the app's cart UI supports multi-vendor carts in a single checkout. Until this is reconciled, AOV and item-count figures computed from the *purchase* event risk mixing single- and multi-item semantics.

Fix: Backend to confirm the real order model in writing before the funnel numbers are used for GMV/AOV reporting to stakeholders.

4. Requirements by Audience

Who needs this data, and in what form, once the collector and dashboard exist.

Team	What they need	Why
Backend	Build the collector endpoint (spec ready); add the ~10 new event types; build the admin/vendor REST API (spec ready); build an aggregation layer feeding the dashboard.	Nothing downstream works until this exists — every other row depends on it.
Stakeholders / Owner	Weekly funnel report (view→cart→checkout→purchase); GMV, AOV, commission revenue; vendor & courier health at a glance.	Currently 100% manual/unavailable — this is the war-room view.
Marketing	Coupon redemption & ROI; guest-vs-logged-in split; notification open rates; (Phase 2) UTM/channel attribution to first view_item.	Promo spend and acquisition channels are currently unmeasured.
Vendor Operations	Vendor funnel (listing→order→commission); commission wallet health & at-risk vendors; product approval queue.	Vendor supply is the marketplace's other half — commission is the revenue proof point.
Delivery Operations	Courier cash-in-hand ledger; delivery SLA (pickup→deliver time); package-pilot funnel; COD collection reliability by city.	Cash exposure and pilot viability are currently invisible.
Finance	GMV reconciliation vs. order DB ($\pm 1\%$ target already agreed); commission collected vs. outstanding; refunds/disputes.	Revenue integrity check independent of the app.

5. Consolidated Event Catalog

The full list the backend collector must accept. **EXISTING** events are already firing from the app today (names are frozen — do not rename). **NEW** events close the gaps in Section 3.

Event		Fired when	Key properties	Primary consumer
view_item	● EXISTING	Product detail loads	item_id, category, seller_id, price_egg, guest	Stakeholders
add_to_cart	● EXISTING	Add-to-cart succeeds	item_id, quantity, cart_value_egg	Stakeholders
view_cart	● NEW	Cart screen opened	cart_value_egg, item_count	Stakeholders
cart_remove	● NEW	Item removed from cart	item_id, cart_value_egg	Stakeholders
begin_checkout	● EXISTING	Checkout screen mounts	cart_value_egg, item_count, vendor_count	Stakeholders
checkout_payment_method_selected	● EXISTING	Payment method picked	method	Marketing
coupon_applied / removed / failed	● NEW	Discount code used at checkout	code, discount_egg, reason	Marketing
checkout_failed / payment_declined	● NEW	Checkout does not complete	reason, step	Stakeholders
purchase	● EXISTING	COD order placed	order_id, value_egg, currency, payment_type, item_count	Stakeholders, Finance
address_location_selected	● NEW	Governorate/city chosen	governorate_id, city_id	Delivery Ops
login_success / register_success / logout	● EXISTING	Auth lifecycle	method, role	Marketing
login_prompt_shown	● EXISTING	Guest hits account gate	screen context	Marketing
screen_view	● EXISTING	Every navigation	screen_name	Stakeholders
search_performed	● EXISTING	Explore search returns results	query, result_count	Marketing
wishlist_add / remove	● EXISTING	Wishlist toggle	item_id	Marketing
listing_published / resubmitted / status_changed / deleted	● EXISTING	Vendor listing lifecycle	item_id, category, price_egg, status	Vendor Ops
order_status_changed	● EXISTING	Any order transition	order_id, status, role, method, reason	Vendor Ops, Delivery Ops

Event		Fired when	Key properties	Primary consumer
commission_threshold_warned / paused / settled	● NEW	Vendor wallet crosses threshold / pays	vendor_id, outstanding_egg, threshold	Vendor Ops, Finance
delivery_request_submitted / priced / confirmed / picked_up / delivered / cancelled	● NEW	Package pilot lifecycle	request_id, price_egg, city_pair	Delivery Ops
courier_cash_handover	● NEW	Courier deposits collected cash	courier_id, amount_egg	Delivery Ops, Finance
notification_received / opened	● NEW	Push notification lifecycle	notification_id, category	Marketing

6. Business Flows

6.1 Core buyer funnel — the app's north-star metric

Red boxes mark diagnostic events that don't exist yet; without them, a drop in conversion cannot be explained.
Guest or signed-in user

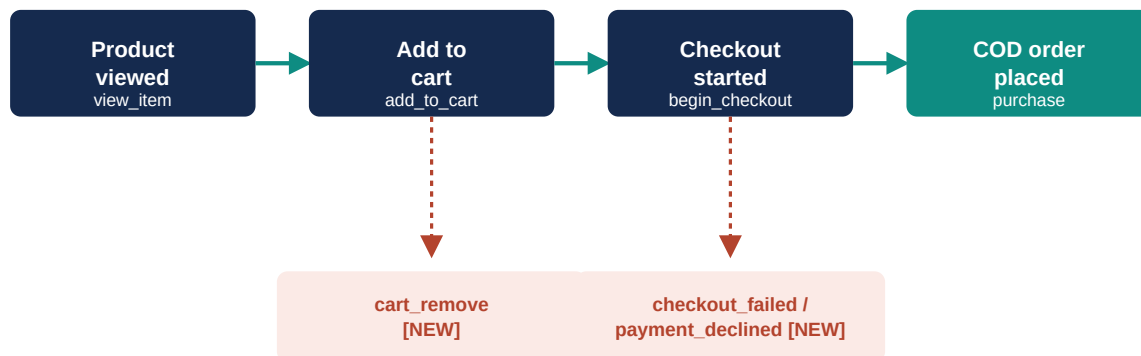
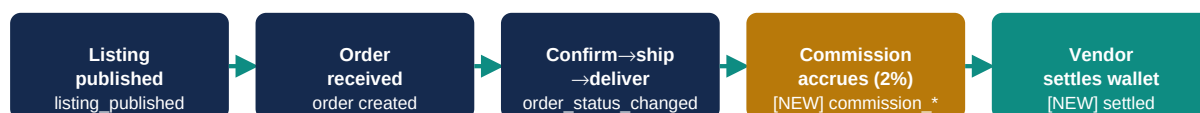


Figure 1 — Guest or signed-in buyer funnel, with missing drop-off diagnostics highlighted.

6.2 Vendor funnel — commission is the revenue proof point



Outstanding $\geq$ warnThreshold $\rightarrow$ vendor notified | $\geq$ pauseThreshold $\rightarrow$ new listings blocked

Figure 2 — Vendor lifecycle from listing to commission settlement. Amber/teal stages are entirely unmeasured today.

6.3 Delivery pilot & cash ledger

"Send Package" pilot — zero analytics today [ALL EVENTS NEW]

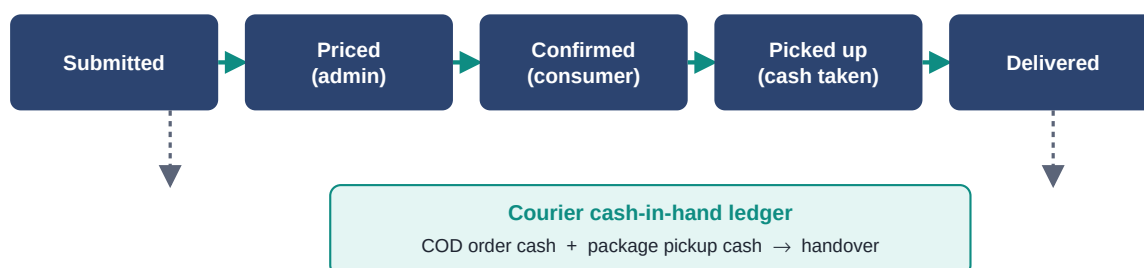
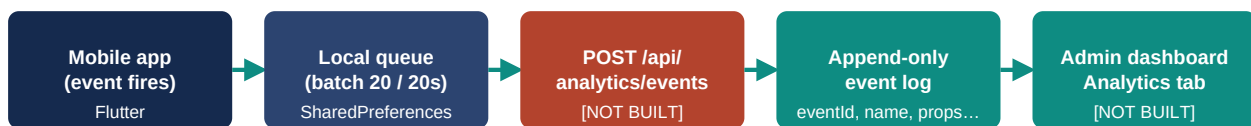


Figure 3 — "Send Package" pilot shares the courier cash-in-hand ledger with COD orders; both need the same handover tracking.

6.4 Data flow — from app event to stakeholder report



Weekly funnel report → Stakeholders | Vendor funnel → Vendor Ops | Courier ledger → Delivery Ops

Figure 4 — The collector endpoint (red) is the single blocker; everything downstream is waiting on it.

7. Dashboard Features Required

The admin dashboard prototype (static HTML, hard-coded data) already specs most screens needed. This section adds what's missing to make the analytics above visible, and flags where each screen's data source is today.

Screen	Must show	Status
Vendor home	7-day sales & orders, orders needing action, low/out-of-stock, commission status	Spec exists; not wired
Admin home	GMV + order volume trend, approval queue size, open disputes, payment mix	Spec exists; not wired
Analytics tab (NEW)	Funnel conversion rates (view→cart→checkout→purchase), guest vs. logged-in split, cohort by governorate, vendor funnel view	Not in current spec — add it
Commission wallet view	Per-vendor outstanding balance, warn/pause thresholds, settlement history	Spec exists; not wired
Courier & delivery pilot view	Cash-in-hand per courier, handover log, package pilot funnel by status	Spec exists; not wired; pilot funnel is new

8. Prioritized Roadmap

P0 — Immediate / pre-launch

- Ship the POST /api/analytics/events collector (spec is ready — no design work left, just build it).
- Add view_cart, cart_remove, checkout_failed / payment_declined, coupon_applied / removed — cheapest fixes, highest diagnostic value for the existing north-star funnel.
- Wire the Analytics tab to real data instead of leaving funnel numbers unavailable to anyone.

P1 — Launch + 30 days

- Instrument commission_* events and the delivery pilot lifecycle end to end.
- Build the admin/vendor REST API must-haves from the existing BACKEND_HANDOFF spec (orders, vendors, commission, couriers).
- Resolve the one-listing-per-order vs. multi-vendor-cart mismatch before GMV/AOV figures go into stakeholder reporting.

P2 — Post-launch

- notification_received / opened for push ROI.

- address_location_selected and governorate cohorting for expansion decisions.
- UTM/channel attribution to first view_item for marketing.

9. Decisions Needed From the Owner

Carried forward from existing docs plus what this review surfaced — none of these block P0 shipping the collector, but they should be settled before P1.

#	Decision	Recommendation
1	Should event properties be a strict typed schema per event, or accepted as opaque JSON?	Opaque JSON for v1 — matches the client today; tighten once dashboard query patterns are known.
2	One-listing-per-order (documented backend model) vs. multi-vendor cart (app UI) — which is correct?	Backend to confirm in writing before AOV/GMV figures are trusted for reporting.
3	Admin dashboard build approach: Flutter Web, off-the-shelf internal tool, or bespoke React?	Vendor side on Flutter Web (reuse app code); admin moderation on an off-the-shelf tool at launch; revisit bespoke admin post-launch.
4	Store-level coupons for vendors, or platform-only coupons?	Platform-only at launch — simpler, smaller abuse surface; revisit once coupon_applied data exists.

10. Source Reference

This document synthesizes and does not duplicate the following, all in the xStore repository:

- lib/core/analytics/analytics_service.dart, event_names.dart, analytics_event.dart — the shipping client pipeline
- docs_business/backend/03_ANALYTICS_EVENTS_HANDOFF.md — wire contract for the collector endpoint
- docs_business/launch_todos/03_funnel_metrics.md — the approved north-star funnel and event names
- docs_business/03_dashboard_product_vision.md — vendor/admin dashboard product vision and role model
- docs_business/admin-dashboard/BACKEND_HANDOFF.md — full admin/vendor REST API spec
- lib/core/network/api_endpoints.dart, delivery_api_endpoints.dart — backend surface area actually integrated